

Unraveling the Relationships between Age, Hearing Loss, Mental Health, and Vection

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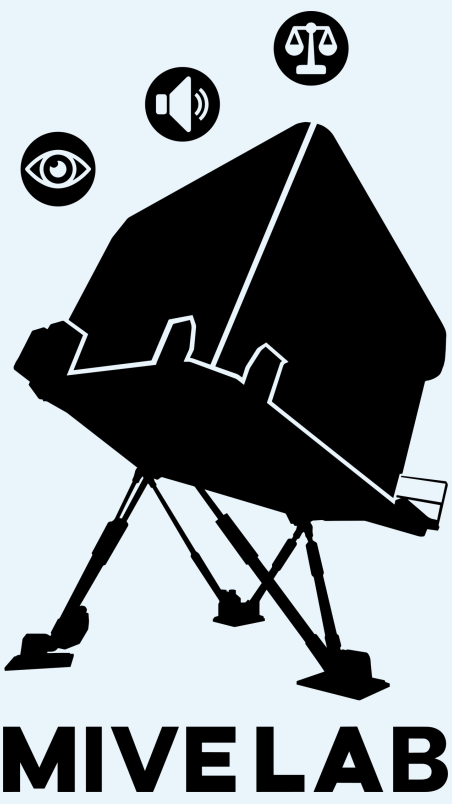


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Background

- A common challenge in aging is that of **age-related hearing loss (ARHL)**. People with ARHL are 3x as likely to experience falls than those without.¹
- The prevalence of anxiety is **40% in those with ARHL**², and ARHL is **positively associated with depression**.³
- Vection is the perception of self-motion in the absence of physical motion and is influenced by visual cues.^{4,5} Older people with ARHL rely more on visual cues.⁶
- Studies show that vection is greater in older adults and those with anxiety but weaker in those with depression. However, these findings are mixed.⁷

What is the relationship between age, ARHL, mental health, and vection?

Methods

Data collection is ongoing but we hope to recruit a total of **90 participants** stratified by **age** and **hearing status**: **30 young adults (controls)**, **30 older adults** (no ARHL; < 20 dB PTAs), and **30 older adults** (ARHL, ≥ 20 dB PTAs).

Pre-Experiment: Baseline Battery & Questionnaires



Audition



Vision



Vestibular



Multisensory



Cognition

Questionnaires include **Dizziness Handicap Inventory (DHI)** and **Hearing Handicap Inventory for the Elderly (HHIE-S)**.

Study Design

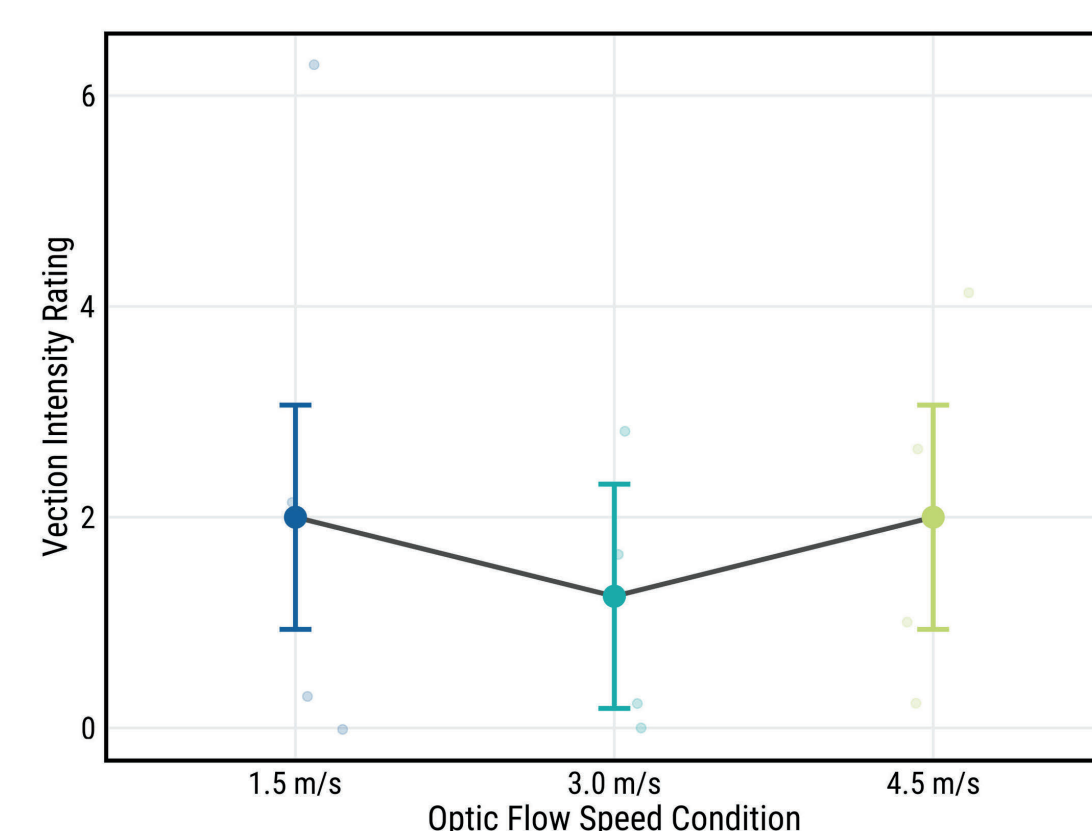


Participants perform a seated vection task in *StreetLab*.



- Continuous forward translational movement** in a virtual grocery store at speeds **1.5 m/s**, **3.0 m/s**, and **4.5 m/s**
- Vection intensity ratings** on a scale of **1 to 10** and onset and duration.

Preliminary Results

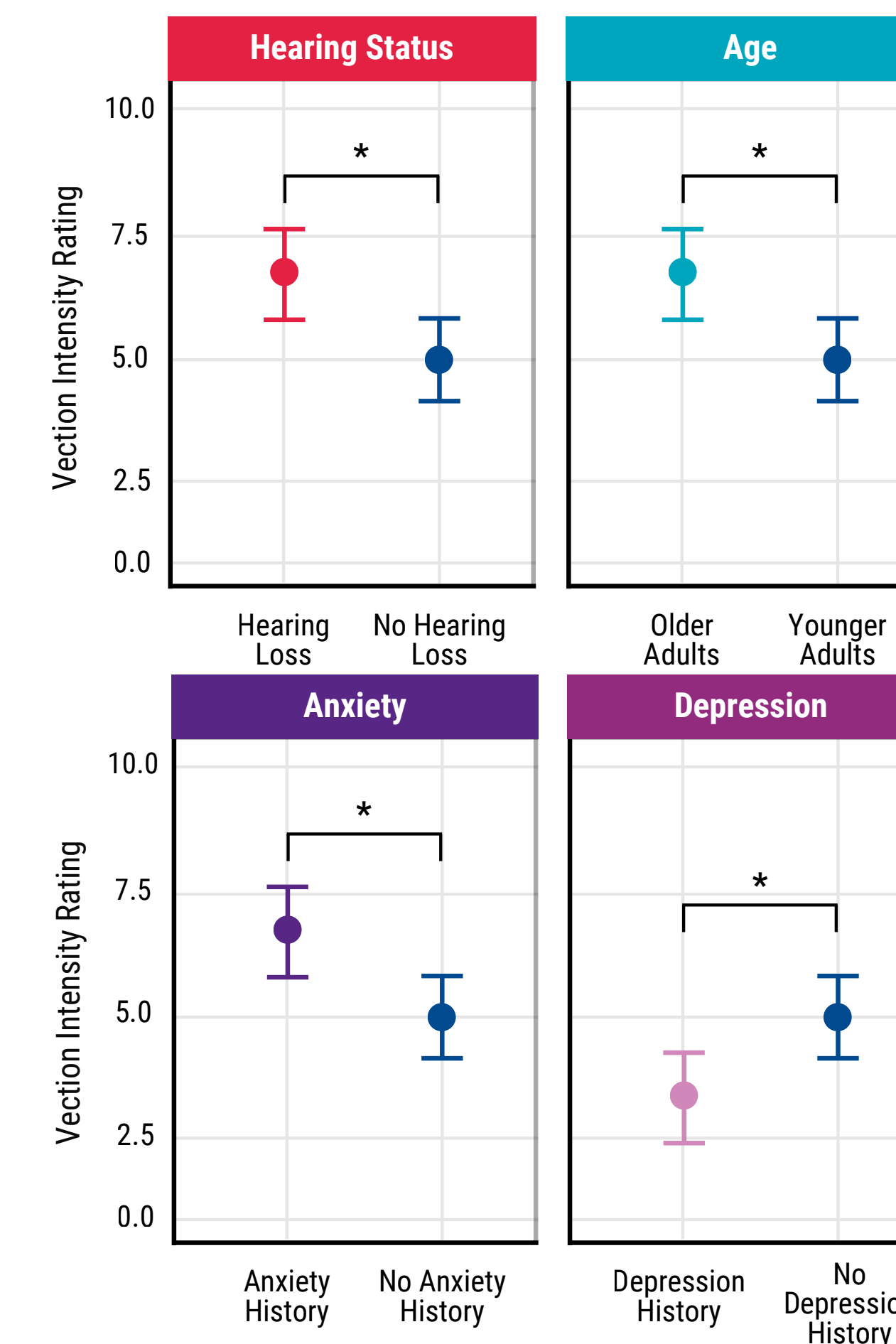


So far, with our young adult sample ($n = 8$), participants are reporting similar vection ratings at **1.5 m/s** and **4.5 m/s** and lower vection ratings at **3.0 m/s speed**. This may be because of perceptual dissimilarity between speeds.

Proposed Analyses

We will run repeated-measures **two ANOVAs**: (i) 3 Condition x 2 Group (Normal Hearing, Hearing Loss) on vection and (ii) 3 Condition x 2 Age Group (YAs, OAs; normal hearing) on vection. We will also run **multi-variate analyses** with mental health history, DHI, HHIE-S, and field dependence as predictors of vection outcomes.

Predicted Results



For all predictions we have regarding vection, we are including effects on not only intensity but also onset and duration. Based on prior findings, we have four key predictions:

Role of Age and Hearing Status

- Older adults with ARHL will report **greater vection** compared to those with normal hearing.
- Older adults will exhibit **greater vection** relative to younger adults.

Role of Mental Health

- Older adults with ARHL and anxiety history will report **greater vection**.
- Older adults with ARHL and depression history will exhibit **reduced vection**.

Limitations & Future Directions

- Lack the capacity to directly tackle questions about mental health as we do not have **standardized measures of depression and anxiety severity**.
- Lack experimental power to examine specific questions about postural control in ARHL and mental health.
- Implement **standardized measures of depression and anxiety** (e.g. BAI, GAD-7, etc) to test mental health hypotheses
- Elucidate postural control biomechanics by **manipulating balance** (i.e. force plate elevations)

References: ¹Iam et al. (2016). *The Laryngoscope*, 126(11), 2587–2596. ²Brewster et al. (2018). *The Am J Geriatr Psychiatry*, 26(7), 788–796. ³Blazer & Tucci. (2019). *Psychol Med*, 49(6), 891–897. ⁴Palmisano et al. (2014). *Exp Brain Res*, 232, 1185–1191. ⁵Tamada & Seno. (2015). *Aerosp Med Hum Perform*, 86(9), 794–802. ⁶Ramkhalwansingh et al. (2018). *Psychol Aging*, 33(5), 798–813. ⁷Murovec, B., Spaniol, J., Campos, J. L., & Keshavarz, B. (2022). *Perception*, 51(10), 698–714.